Applied Mathematics 205

Unit 0. Introduction

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Scientific Computing

Computation is now recognized as the "third pillar" of science (along with theory and experiment). Why?

- Practically relevant mathematical models do not have analytical solutions
- Large amounts of data need to be processed automatically
- Modern computers can handle large-scale problems

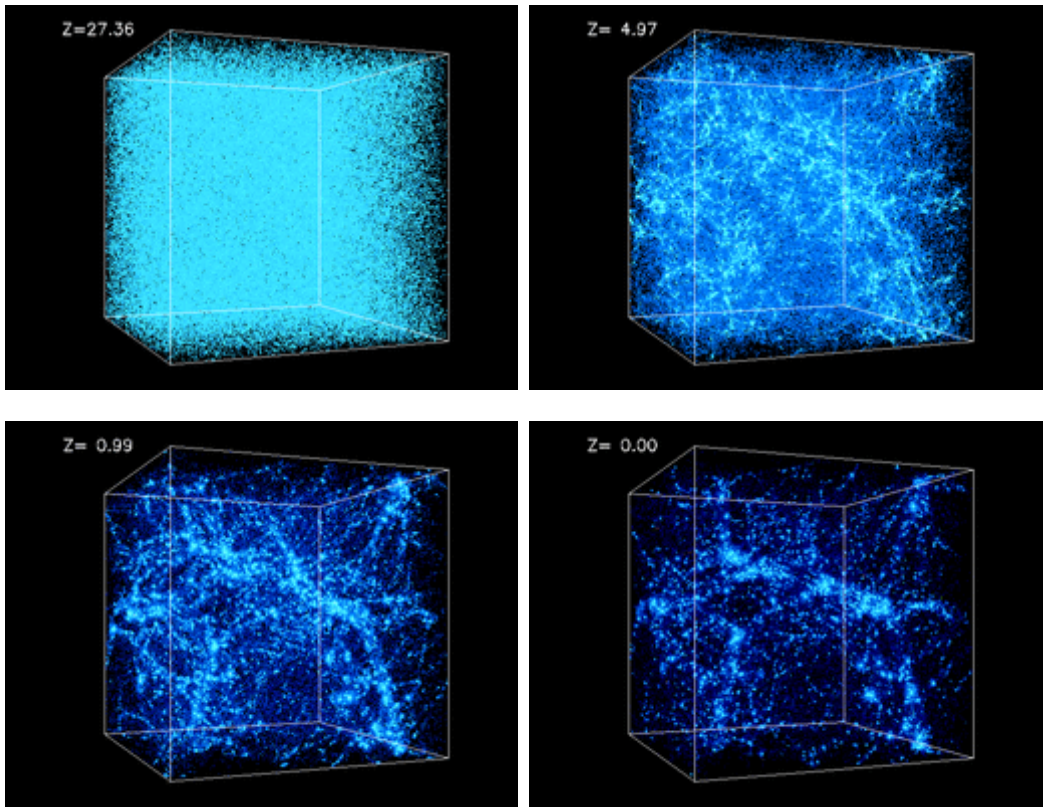
What is Scientific Computing

- Scientific computing is closely related to numerical analysis
- *"Numerical analysis is the study of algorithms for the problems of continuous mathematics"* [Nick Trefethen, SIAM News, 1992]
- **Continuous mathematics** involves real numbers as opposed to integers
- **Numerical analysis** studies these algorithms
- **Scientific computing** applies them to practical problems
- **Scientific computing** is distinct from Computer Science, which focuses on discrete mathematics (e.g. graph theory)

Applications of Scientific Computing

Cosmology

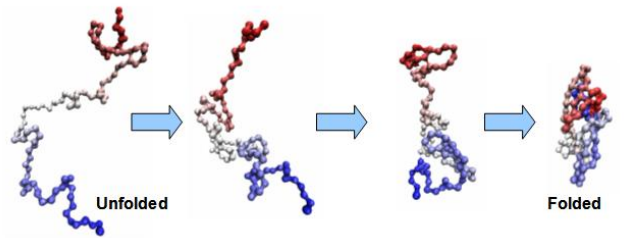
Cosmological simulations to test theories of galaxy formation



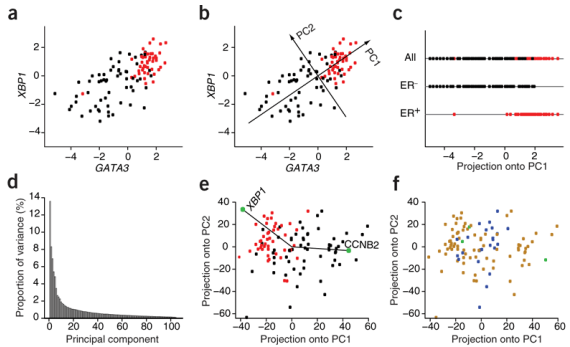
http://cosmicweb.uchicago.edu/filaments.html

Biology

- Protein folding

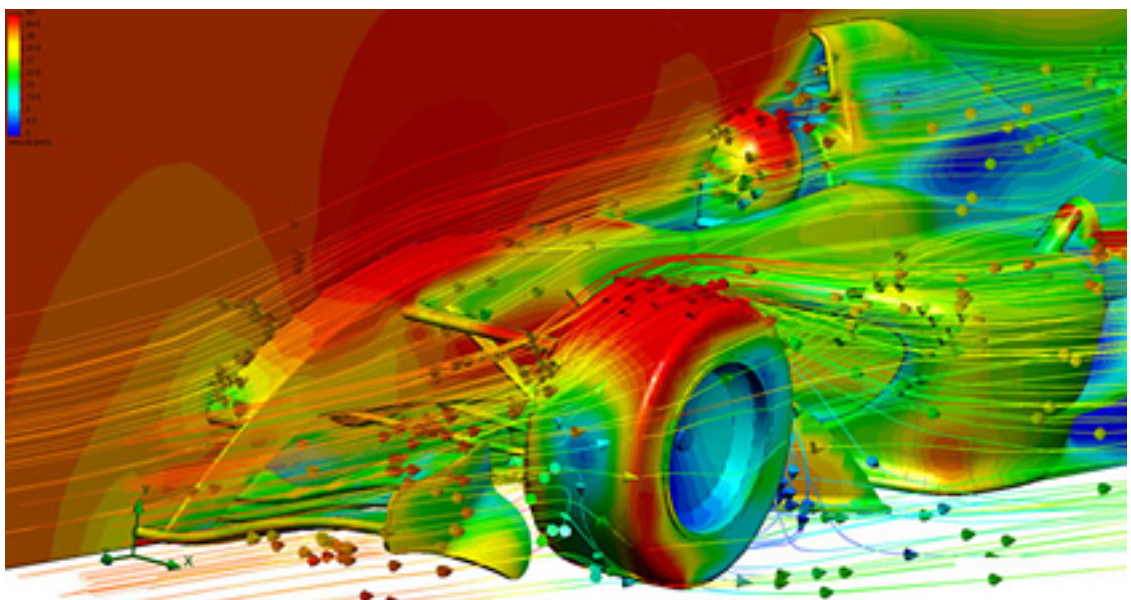


- Statistical analysis of gene expression



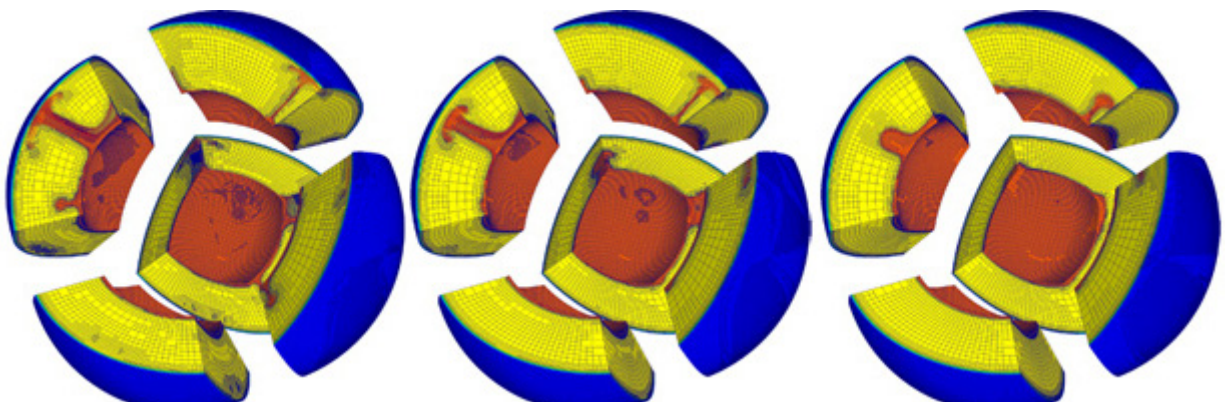
Computational Fluid Dynamics

- CFD simulations replace or complement wind-tunnel experiments
- Computational geometry is easier to tweak than a physical model
- Simulations provide the entire flow field, not available experimentally



Geophysics

- Experimental data is only available on Earth's surface
- Simulations help to test models of the interior



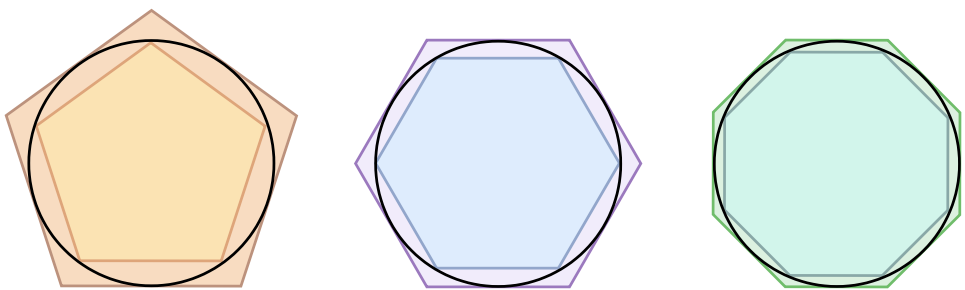
Calculation of π

Calculation of π

- π is the ratio of a circle's circumference to its diameter
- Babylonians (1900 BC): 3.125
- From the Old Testament (1 Kings 7:23):
  "And he made the molten sea of ten cubits from brim to brim, round in compass, and the height thereof was five cubits; and a lines of thirty cubits did compass it round about"
  Implies $\pi \approx 3$
- Egyptians (1850 BC): $(\frac{16}{9})^2 \approx 3.16$

Calculation of π

- Archimedes (287-212 BC) bounded π by perimeters of regular polygons: inscribed and superscribed



- For 96-sided polygon: $\frac{223}{71} < \pi < \frac{22}{7}$ (interval length: 0.00201)
- Example of an infinite process converging to the exact solution
- Provides both the estimate **and** error bounds

Calculation of π

- James Gregory (1638-1675) discovers the arctangent series

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$$

- Setting $x = 1$ gives

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$$

  but it converges very slowly

Calculation of π

- The arctangent series converges faster for points closer to 0

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$$

- John Machin (1680-1752) observed that

$$\frac{\pi}{4} = 4 \arctan \frac{1}{5} - \arctan \frac{1}{239}$$

  and computed 100 digits of π
- Derivation
  - $\tan \alpha = \frac{1}{5}$
  - $\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha} = \frac{5}{12}$
  - $\tan 4\alpha = \frac{2 \tan 2\alpha}{1 - \tan^2 2\alpha} = \frac{120}{119}$
  - $\tan \left(4\alpha - \frac{\pi}{4}\right) = \frac{\tan 4\alpha - 1}{1 + \tan 4\alpha} = \frac{1}{239}$

Calculation of π

Users of Manchin's formula

1706	John Machin	100 digits
1719	Thomas de Lagny	112 digits
1739	Matsunaga Ryohitsu	50 digits
1794	Georg von Vega	140 digits
1844	Zacharias Dase	200 digits
1847	Thomas Clausen	248 digits
1853	William Rutherford	440 digits
1876	William Shanks	707 digits

Sources of Error in Scientific Computing

Sources of Error in Scientific Computing

- There are several sources of error in solving real-world problems
- Some are beyond our control
  (e.g. uncertainty in modeling parameters or initial conditions)
- Some are introduced by our numerical approximations
  - Truncation/discretization error:
    Objects from continuous mathematics need to be discretized
    (finite differences, truncated infinite series...)
  - Rounding error:
    Computers work with finite precision arithmetic

Sources of Error in Scientific Computing

- It is crucial to understand and control the error introduced by numerical approximation, otherwise the results might be garbage
- This is a major part of Scientific Computing, called error analysis
- Error analysis becomes more important for larger scale problems as errors accumulate
- Most people are familiar with rounding error,
  but discretization error is far more important in practice

Discretization Error vs. Rounding Error

- Consider a finite difference approximation to $f'(x)$

$$f_{\text{diff}}(x;h) = \frac{f(x+h) - f(x)}{h}$$

- From Taylor series (for $\theta \in [x, x+h]$)

$$f(x+h) = f(x) + hf'(x) + f''(\theta)h^2/2$$

  we see that

$$f_{\text{diff}}(x;h) = \frac{f(x+h) - f(x)}{h} = f'(x) + f''(\theta)h/2$$

- Suppose $|f''(\theta)| \leq M$, then bound on discretization error is

$$|f'(x) - f_{\text{diff}}(x;h)| \leq Mh/2$$

Discretization Error vs. Rounding Error

- But we can't compute $f_{\text{diff}}(x;h)$ in exact arithmetic
- Let $\tilde{f}_{\text{diff}}(x;h)$ denote finite precision approximation of $f_{\text{diff}}(x;h)$
- Numerator of $\tilde{f}_{\text{diff}}$ introduces rounding error $\lesssim \epsilon|f(x)|$
  (on modern computers $\epsilon \approx 10^{-16}$, will discuss this shortly)
- Hence we have the rounding error

$$\left|f_{\text{diff}}(x;h) - \tilde{f}_{\text{diff}}(x;h)\right| \lesssim \left|\frac{f(x+h)-f(x)}{h} - \frac{f(x+h)-f(x)+\epsilon f(x)}{h}\right| = \epsilon|f(x)|/h$$

Discretization Error vs. Rounding Error

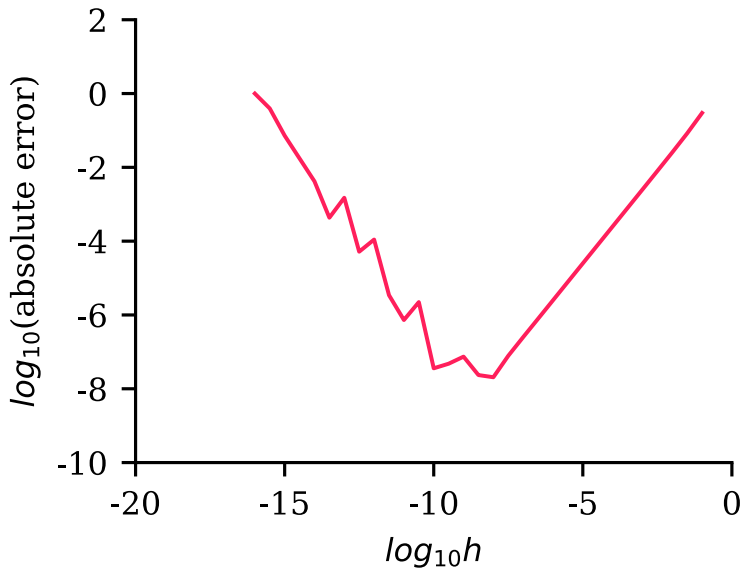
- Then we can bound the total error (discretization and rounding)

$$
\begin{aligned}
|f'(x) - \tilde{f}_{\text{diff}}(x;h)| &= |f'(x) - f_{\text{diff}}(x;h) + f_{\text{diff}}(x;h) - \tilde{f}_{\text{diff}}(x;h)| \\
&\leq |f'(x) - f_{\text{diff}}(x;h)| + |f_{\text{diff}}(x;h) - \tilde{f}_{\text{diff}}(x;h)| \\
&\leq Mh/2 + \epsilon|f(x)|/h
\end{aligned}
$$

- Since ϵ is so small, here we expect discretization error to dominate until h gets sufficiently small

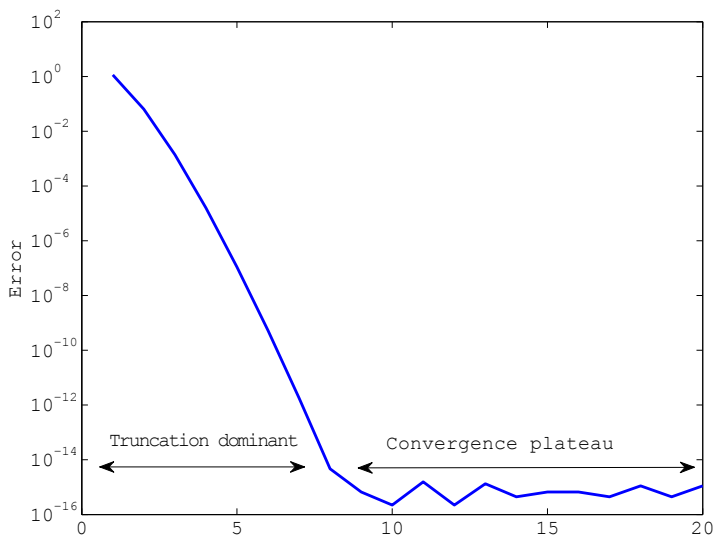
Discretization Error vs. Rounding Error

- Consider $f(x) = \exp(5x)$.
  Error of $f_{\text{diff}}(x, h)$ at $x = 1$ as function of h



Discretization Error vs. Rounding Error

- Note that in the finite difference example,
  we observe error growth due to rounding as $h \to 0$
- A more common situation (that we'll see in Unit 1, for example)
  is that the error plateaus at around ϵ due to rounding error



Absolute vs. Relative Error

- Recall our bound $|f'(x) - \tilde{f}_{\text{diff}}(x;h)| \leq Mh/2 + \epsilon|f(x)|/h$
- This is a bound on Absolute Error

$$\text{Absolute Error} = \text{true value} - \text{approximate value}$$

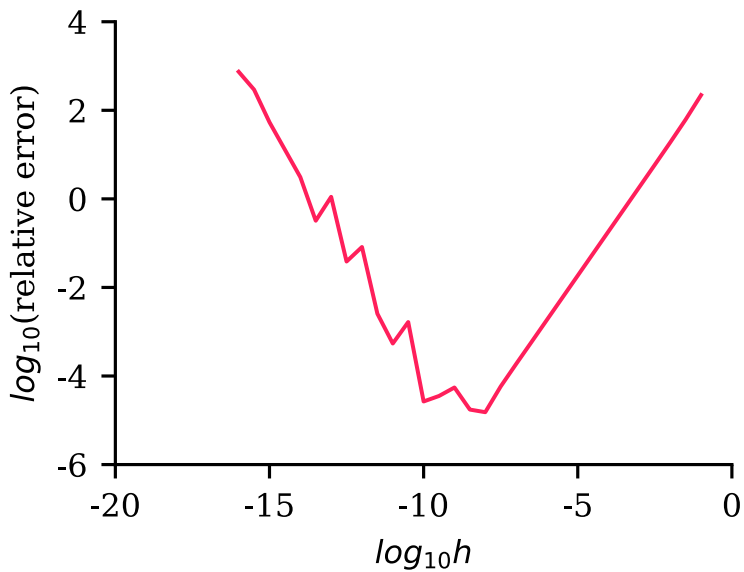
- Generally more interesting to consider Relative Error

$$\text{Relative Error} \equiv \frac{\text{Absolute Error}}{\text{true value}}$$

- Relative error is a dimensionless quantity
- If unknown, true value is replaced with an estimate

Absolute vs. Relative Error

- For our finite difference example, plotting relative error just rescales the error values



Convergence Plots

- We have shown several plots of error as a function of a discretization parameter
- In general, these plots are very important in scientific computing to demonstrate that a numerical method is behaving as expected
- To display convergence data in a clear way, it is important to use appropriate axes for our plots

Convergence Plots

- Most often we will encounter algebraic convergence, where error decreases as Ch^q for some $C, q \in \mathbb{R}$
- Algebraic convergence: If $E = Ch^q$, then

$$\log E = \log C + q \log h$$

- Plotting algebraic convergence on log–log axes asymptotically yields a straight line with slope q
- Hence a good way to deduce the algebraic convergence rate is by comparing error to Ch^q on log–log axes

Convergence Plots

- Sometimes we will encounter exponential convergence, where error decays as Ce^{-qN} as $N \to \infty$
- If $E = Ce^{-qN}$ then

$$\log E = \log C - qN$$

- Hence for exponential convergence, better to use log-linear axes (like the previous "error plateau" plot)

Numerical Sensitivity

- In practical problems we will always have input perturbations (modeling uncertainty, rounding error)
- Let $y = f(x)$, and denote perturbed input $\hat{x} = x + \Delta x$
- Also, denote perturbed output by $\hat{y} = f(\hat{x})$, and $\hat{y} = y + \Delta y$
- The function f is *sensitive* to input perturbations if $\Delta y \gg \Delta x$
- This is sensitivity inherent in f, independent of any approximation (though a numerical approximation $\hat{f} \approx f$ can exacerbate sensitivity)

Sensitivity and Conditioning

- For a sensitive problem,
  small input perturbation leads to large output perturbation
- Can be made quantitative with the concept of (relative) condition number

$$\text{Condition number} = \frac{|\Delta y/y|}{|\Delta x/x|}$$

- Problems with Condition number $\gg 1$ are called ill-conditioned.
  In such problems, small input perturbations are amplified

Sensitivity and Conditioning

- Condition number can be analyzed for various problem types (independent of algorithm used to solve the problem). Examples:
  - Function evaluation, $y = f(x)$
  - Matrix multiplication, $Ax = b$ (solve for b given x)
  - Linear system, $Ax = b$ (solve for x given b)

Conditioning: Function Evaluation

- Problem: evaluate function, $y = f(x)$
- Perturbed problem: $y + \Delta y = f(x + \Delta x)$
- Change in x: Δx
- Change in y: $\Delta y \approx f'(x)\Delta x$
- Condition number is the ratio of relative changes

$$\kappa = \frac{f'(x)\Delta x / f(x)}{\Delta x / x} = \frac{f'(x)x}{f(x)}$$

Conditioning: Matrix Multiplication

- Problem: multiply matrix and vector, $b = Ax$
- Perturbed problem: $b + \Delta b = A(x + \Delta x) \implies \Delta b = A\Delta x$
- Condition number is

$$\kappa = \frac{\|\Delta b\|/\|b\|}{\|\Delta x\|/\|x\|} = \frac{\|A\Delta x\|}{\|\Delta x\|}\,\frac{\|x\|}{\|Ax\|} = \frac{\|A\Delta x\|}{\|\Delta x\|}\,\frac{\|A^{-1}b\|}{\|b\|}$$

- Matrix norm

$$\|A\| = \max_{v \neq 0} \frac{\|Av\|}{\|v\|}$$

- Condition number $\kappa(A)$ from linear algebra is an upper bound for κ

$$\kappa = \frac{\|A\Delta x\|}{\|\Delta x\|}\,\frac{\|A^{-1}b\|}{\|b\|} \leq \|A\|\,\|A^{-1}\| = \kappa(A)$$

Conditioning: Linear System

- Problem: solve linear system $Ax = b$
- Perturbed problem: $A(x + \Delta x) = b + \Delta b \implies A\Delta x = \Delta b$
- Condition number is

$$\kappa = \frac{\|\Delta x\|/\|x\|}{\|\Delta b\|/\|b\|} = \frac{\|\Delta x\|}{\|A\Delta x\|}\,\frac{\|Ax\|}{\|x\|} = \frac{\|A^{-1}\Delta b\|}{\|\Delta b\|}\,\frac{\|Ax\|}{\|x\|}$$

- Matrix norm

$$\|A\| = \max_{v \neq 0} \frac{\|Av\|}{\|v\|}$$

- Condition number $\kappa(A)$ from linear algebra is an upper bound for κ

$$\kappa = \frac{\|A^{-1}\Delta b\|}{\|\Delta b\|}\,\frac{\|Ax\|}{\|x\|} \leq \|A^{-1}\|\,\|A\| = \kappa(A)$$

Exercise: Diagonal Matrix

$$A = \begin{pmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{pmatrix}$$

- Matrix norm

$$\|A\| = \max_{v \neq 0} \frac{\|Av\|}{\|v\|} = \max(|d_1|, |d_2|, |d_3|)$$

- Condition number

$$\kappa(A) = \|A\| \|A^{-1}\| = \frac{\max(|d_1|, |d_2|, |d_3|)}{\min(|d_1|, |d_2|, |d_3|)}$$

Stability of an Algorithm

- In practice, we solve problems by applying a numerical method to a mathematical problem, e.g. apply Gaussian elimination to $Ax = b$
- To obtain an accurate answer, we need to apply a stable numerical method to a well-conditioned mathematical problem
- Question: What do we mean by a stable numerical method?
- Answer: Roughly speaking, the numerical method doesn't accumulate error (e.g. rounding error) and produce garbage
- We will make this definition more precise shortly, but first, we discuss rounding error and finite-precision arithmetic

Code Examples

- From here on, a number of code examples will be provided
- They will be available via Git repository
  code.harvard.edu/AM205/public
- Git is one example of version control software, which tracks the changes of files in a software project. Features:
  - Compare files to any previous version
  - Merge changes in the same files by multiple people
  - Not suitable for binary files (Word, PDF, images, videos, etc)
- Note: Avoid storing large binary files in repositories.
  They cannot be removed without rewriting history

Git

- Git can be installed as a command-line utility on all major systems.
- For authentication, you will need to add an SSH key to your profile at code.harvard.edu
- Follow this guide to generate an SSH key
- To download a copy of the repository, use

```
git clone git@code.harvard.edu:AM205/public.git
```

- Then, at later times, you can type

```
git pull
```

to obtain any updated files. Graphical interfaces for Git are also available

Finite-Precision Arithmetic

- Key point: we can only represent a finite and discrete subset of the real numbers on a computer.
- The standard approach in modern hardware is to use binary floating point numbers (basically "scientific notation" in base 2),

$$
\begin{aligned}
x &= \pm(1 + d_1 2^{-1} + d_2 2^{-2} + \ldots + d_p 2^{-p}) \times 2^E \\
&= \pm(1.d_1 d_2 \ldots d_p)_2 \times 2^E
\end{aligned}
$$

Finite-Precision Arithmetic

- We store

$$\underbrace{\pm}_{\text{1 sign bit}} \qquad \underbrace{d_1, d_2, \ldots, d_p}_{p \text{ mantissa bits}} \qquad \underbrace{E}_{\text{exponent bits}}$$

- Note that the term bit is a contraction of "binary digit"
- This format assumes that $d_0 = 1$ to save a mantissa bit, but sometimes $d_0 = 0$ is required, such as to represent zero.
- The exponent resides in an interval $L \leq E \leq U$.

IEEE Floating Point Arithmetic

- Universal standard on modern hardware is IEEE floating point arithmetic (IEEE 754), adopted in 1985
- Development led by Prof. William Kahan (UC Berkeley), who received the 1989 Turing Award for his work

	total bits	p	L	U
IEEE single precision	32	23	-126	127
IEEE double precision	64	52	-1022	1023

- Note that single precision has 8 exponent bits but only 254 (not 256) different values of E, since some exponent bits are reserved to represent special numbers

Exceptional Values

- These exponents are reserved to indicate special behavior, including values such as Inf and NaN:
  - Inf = "infinity", e.g. $1/0$ (also $-1/0 = -\text{Inf}$)
  - NaN = "not a number", e.g. $0/0$, Inf/Inf

IEEE Floating Point Arithmetic

- Let $\mathbb{F}$ denote the floating point numbers. Then $\mathbb{F} \subset \mathbb{R}$ and $|\mathbb{F}| < \infty$.
- Question: How should we represent a real number x, which is not in $\mathbb{F}$?
- Answer: There are two cases to consider:
  - Case 1: x is outside the range of $\mathbb{F}$ (too small or too large)
  - Case 2: The mantissa of x requires more than p bits.

IEEE Floating Point Arithmetic

Case 1: x is outside the range of $\mathbb{F}$ (too small or too large)

- Too small:
  - Smallest positive value that can be represented in double precision is $\approx 10^{-323}$
  - For a value smaller than this we get underflow, and the value typically set to 0
- Too large:
  - Largest $x \in \mathbb{F}$ ($E = U$ and all mantissa bits are 1) is approximately $2^{1024} \approx 10^{308}$
  - For values larger than this we get overflow, and the value typically gets set to Inf

IEEE Floating Point Arithmetic

Case 2: The mantissa of x requires more than p bits

- Need to round x to a nearby floating point number
- Let $\texttt{round} : \mathbb{R} \to \mathbb{F}$ denote our rounding operator
- There are several different options:
  round up, round down, round to nearest, etc
- This introduces a rounding error
  - absolute rounding error $x - \texttt{round}(x)$
  - relative rounding error $(x - \texttt{round}(x))/x$

Machine Precision

- It is important to be able to quantify this rounding error — it's related to machine precision, often denoted as ϵ or ϵ_{mach}
- ϵ is the difference between 1 and the next floating point number after 1, therefore $\epsilon = 2^{-p}$
- In IEEE double precision, $\epsilon = 2^{-52} \approx 2.22 \times 10^{-16}$

Rounding Error

- Let $x = (1.d_1 d_2 \ldots d_p d_{p+1})_2 \times 2^E \in \mathbb{R}_+$.
- Then $x \in [x_-, x_+]$ for $x_-, x_+ \in \mathbb{F}$, where $x_- = (1.d_1 d_2 \ldots d_p)_2 \times 2^E$ and $x_+ = x_- + \epsilon \times 2^E$.
- $\texttt{round}(x) = x_-$ or x_+ depending on the rounding rule, and hence $|\texttt{round}(x) - x| < \epsilon \times 2^E$
- Also, $|x| \geq 2^E$

Rounding Error

- Hence we have a relative error of less than ϵ

$$\left|\frac{\texttt{round}(x) - x}{x}\right| < \epsilon$$

- Another standard way to write this is

$$\texttt{round}(x) = x\left(1 + \frac{\texttt{round}(x) - x}{x}\right) = x(1+\delta)$$

where $\delta = \frac{\texttt{round}(x)-x}{x}$ and $|\delta| < \epsilon$

- Hence rounding gives the correct answer to within a factor of $1 + \epsilon$

Floating Point Operations

- An arithmetic operation on floating point numbers is called a "floating point operation": $\oplus$, $\ominus$, $\otimes$, $\oslash$ versus $+$, $-$, $\times$, $/$
- Computer performance is often measured in Flop/s:
  number of Floating Point OPerations per second
- Supercomputers are ranked based on number of flops achieved in the LINPACK test, which solves dense linear algebra problems
- Currently, the fastest computers are in the 100 petaflop range:
  1 petaflop $= 10^{15}$ floating point operations per second

Supercomputers

See www.top500.org for an up-to-date list of the fastest supercomputers

TOP500 LIST - JUNE 2022

R_{max} and R_{peak} values are in PFlop/s. For more details about other fields, check the TOP500 description.

R_{peak} values are calculated using the advertised clock rate of the CPU. For the efficiency of the systems you should take into account the Turbo CPU clock rate where it applies.

← 1-100 101-200 201-300 301-400 401-500 →

Rank	System	Cores	Rmax (PFlop/s)	Rpeak (PFlop/s)	Power (kW)
1	**Frontier** - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE DOE/SC/Oak Ridge National Laboratory United States	8,730,112	1,102.00	1,685.65	21,100
2	**Supercomputer Fugaku** - Supercomputer Fugaku, A64FX 48C 2.2GHz, Tofu interconnect D, Fujitsu RIKEN Center for Computational Science Japan	7,630,848	442.01	537.21	29,899
3	**LUMI** - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE EuroHPC/CSC Finland	1,110,144	151.90	214.35	2,942
4	**Summit** - IBM Power System AC922, IBM POWER9 22C 3.07GHz, NVIDIA Volta GV100, Dual-rail Mellanox EDR Infiniband, IBM DOE/SC/Oak Ridge National Laboratory United States	2,414,592	148.60	200.79	10,096

R_{max} is from LINPACK, R_{peak} is from clock rate

Supercomputers

Modern supercomputers are very large, link many processors together with fast interconnect to minimize communication time

Frontier at Oak Ridge is 1102 PFlop/s

Floating Point Operation Error

- IEEE standard guarantees that for $x, y \in \mathbb{F}$

$$x \circledast y = \texttt{round}(x * y)$$

  where $*$ and $\circledast$ represent one of the four arithmetic operations
- Hence from our discussion of rounding error, it follows that for $x, y \in \mathbb{F}$

$$x \circledast y = (x * y)(1 + \delta)$$

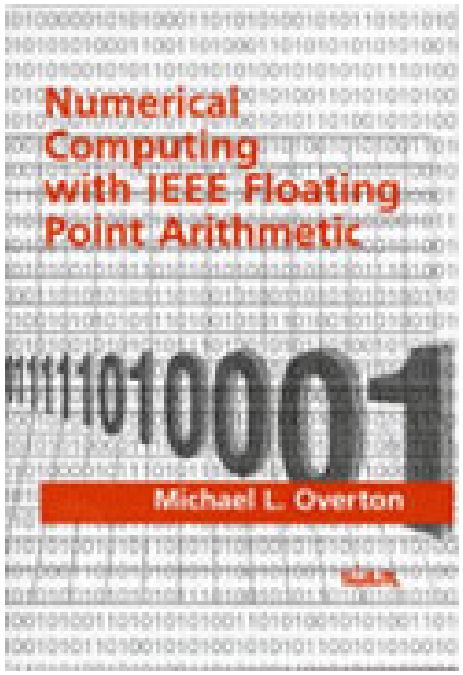
  for some $|\delta| < \epsilon$

Loss of Precision

- Machine precision can be tested. See [unit0/precision.py]
- Since ϵ is so small, we typically lose very little precision per operation

IEEE Floating Point Arithmetic

For more detailed discussion of floating point arithmetic, see:



Michael L. Overton. *Numerical Computing with IEEE Floating Point Arithmetic*. SIAM, 2001 10.1137/1.9780898718072

Numerical Stability of an Algorithm

- We have discussed rounding for a single operation, but in AM205 we will study numerical algorithms that require many operations
- For an algorithm to be useful, it must be stable in the sense that rounding errors do not accumulate and result in "garbage" output
- More precisely, numerical analysts aim to prove backward stability: The method gives the exact answer to a slightly perturbed problem
- For example, a numerical method for solving $Ax = b$ should give the exact answer for $(A + \Delta A)x = (b + \Delta b)$ for small ΔA, Δb

Numerical Stability of an Algorithm

- We note the importance of conditioning:
  backward stability doesn't help us if the mathematical problem is ill-conditioned
- For example, if A is ill-conditioned then a backward stable algorithm for solving $Ax = b$ can still give large error for x
- Backward stability analysis is a deep subject which we do not cover in detail in AM205
- We will, however, compare algorithms with different stability properties and observe the importance of stability in practice